

The Wallet Without a Polity: Displaced and Stateless Subjects under a Bonded Settlement Primitive

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Abstract

The bonded settlement primitive of the mechanism paper, implemented and verified in the implementation paper, exhibits a property whose consequences are most consequential not for populations conventionally treated by labor or commercial law, but for those who occupy the negative space of those frames: refugees, stateless persons, undocumented migrants, and populations whose residency is beyond the reach of stable banking and enforcement infrastructure. The primitive's precondition is a cryptographic key, not a recognized civil-legal subjecthood. This paper develops the consequence.

The wallet-as-legal-subject argument that underwrites the companion labor-law paper extends here in a different direction. In the labor setting the wallet collapses the Roman *res/persona* distinction so that the productive asset is simultaneously property and party; in the present setting it allows the productive asset to act as party where civil-legal subjecthood is itself absent. We do not claim that the primitive resolves statelessness. Arendt [1951] named the right to have rights as the fundamental human right — a right whose denial is a civilizational failure rather than an individual misfortune — and the technological infrastructure considered here cannot grant it. What it grants is narrower: a capacity to have commerce, where the capacity to have rights is denied. The two should not be confused. For populations for whom the first is denied, the second is non-trivial.

The argument is positional rather than programmatic. Real constraints — key custody, counterparty willingness, fiat on-ramp, collateral access — are named and not dismissed. The analytical claim is that the primitive moves the humanitarian-economics gap from an architectural impossibility to an operational engineering question, and that this is a consequential move worth scholarship.

Keywords: statelessness, refugee economies, displaced populations, the wallet as legal subject, Arendt, humanitarian economics, commerce-without-jurisdiction

1 Introduction

The legal infrastructure of commerce presumes a recognized civil subject. Contract law requires parties whose standing the forum will accept; banking access requires identity documents the domestic regime issues; commercial-dispute adjudication requires a court whose jurisdiction reaches both sides. Each of these presumptions is so deeply embedded in the architecture of contemporary trade that its failure mode — populations for whom the presumptions do not hold — is treated by the architecture as exception rather than as a system property.

The United Nations Refugee Agency estimated 120 million forcibly displaced persons worldwide at the end of 2023, and approximately 4.4 million persons registered as stateless, with a

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substantially larger estimated population whose statelessness is undocumented [UNHCR, 2024]. An overlapping and considerably larger population lacks stable banking and enforcement access for reasons of informal migration status, residence without papers, or residency in jurisdictions whose financial infrastructure does not reach them. Across these categories, the productive capacity of well over a hundred million working-age adults is systematically underused because the legal scaffold for engaging it is absent. The humanitarian-economics literature on refugee economies [Betts and Collier, 2017] and the policy frontier between camp and labor market [Pritchett, 2006] documents the cost of this gap and the existing repertoire of responses.

This paper argues that the bonded settlement primitive of the mechanism paper, implemented and verified in the implementation paper, changes the architectural posture of the gap. The primitive operates on parties whose existence as parties does not require civil-legal recognition. A wallet can sign; a bonded commitment can settle; an exchange can complete — without a host-state court, a domestic bank, or a permission-to-work visa sitting between the parties. The labor-law consequences of the primitive are developed in the companion labor-law paper; the present paper develops the consequence for populations to whom the labor-law debate is a second-order question, because the first-order question is whether they are recognized parties to any contract in any forum at all.

What this paper is not. This paper does not claim that the primitive resolves statelessness. It does not confer citizenship, restore legal recognition, or create enforcement against a state that denies a person’s standing. It is not a substitute for the political work that addresses those gaps — work that remains necessary, that this paper does not duplicate, and that no technological infrastructure can perform. It does not predict adoption rates, endorse any particular humanitarian program, or assess the political consequences of widespread wallet adoption among displaced populations. Its argument is structural: the architectural posture of the commerce-access gap changes once a settlement primitive whose precondition is a cryptographic key exists.

Paper organization. Section 2 summarizes the settlement primitive in the terms relevant here. Section 3 restates the wallet-as-legal-subject argument from the labor-law paper in the form needed for the present paper. Section 4 develops the central claim across four moves: scale of the affected population, the architecture they face, what the primitive does and does not do, and the Arendtian distinction between the right to have rights and the capacity to have commerce. Section 5 names the operational constraints the primitive does not dissolve. Section 6 concludes.

2 The Settlement Primitive

We summarize only the features relevant to the present argument. The mechanism is established in the mechanism paper; the Solidity implementation and formal verification in the implementation paper.

A transaction between two parties is consummated by dual-signed EIP-712 commitments. Each party signs from their own wallet. Each party locks collateral against their own performance — the buyer locks $2P$, the seller locks $2G$ where G is the cumulative upstream value bonded in the process. Only the buyer can trigger resolution. Defection is structurally more expensive than cooperation. There is no platform between the parties, no arbitrator authorized to resolve disputes on-chain, and no admin key permitting external override.

The features that matter for the present paper are three. First, the kernel does not require either party to be a recognized civil subject; it requires them to control a key that can sign. Second, the kernel does not require a forum to enforce the agreement; the collateral structure is the enforcement mechanism. Third, the kernel does not route value through a banking intermediary; settlement is direct, wallet to wallet, atomic on resolution.

3 The Wallet as Legal Subject (Brief)

Roman private law distinguished *res* (things, property, objects that can be owned and alienated) from *persona* (persons, parties, subjects capable of acting legally). The distinction survived and structures modern private law. The companion labor-law paper (§4) develops in detail how the bonded primitive collapses the *persona/res* partition: a wallet is simultaneously a property (a key-pair owned and transferable) and a party (a signer, bonder, settlement-receiver). The collapse is operational rather than metaphorical; it follows from the architecture of dual-signed bonded commitments.

For the present paper the relevant consequence is narrower than the labor-law one but harder. In the labor setting the *persona/res* collapse dissolves the subordination gradient between employer and employee, because each party signs for itself and bonds for its own performance. In the present setting — where one or both parties may lack stable *persona* status in the host jurisdiction at all — the collapse means something stronger: the wallet provides a party-to-the-transaction whose existence does not depend on the prior existence of a recognized civil subject. The labor-law argument shows that the wallet can be both property and party. The argument here is that the wallet can be a party where no civil-legal party exists.

This is not a claim that the wallet *is* a civil-legal subject. Civil-legal subjecthood is conferred by politics, and its absence is a political condition. The claim is narrower: a party to a bonded commitment can transact, settle, and accumulate verifiable performance history without first acquiring civil-legal subjecthood, because the kernel does not consult the polity for permission to settle a bond.

4 The Capacity to Have Commerce

4.1 Scale

Beyond the UNHCR figures cited in Section 1, the population relevant to the present argument is broader than the forcibly-displaced and stateless categories alone. Undocumented migrants who hold no recognized status in their country of residence, internally displaced persons whose host sub-jurisdiction does not extend banking infrastructure to their location, and populations in failed-state or contested-sovereignty zones whose nominal citizenship is not exercisable for commercial purposes all share the operationally-relevant property: they cannot reliably enter the legal infrastructure of commerce as recognized parties. Estimates of the combined population vary substantially with definition; the order of magnitude is hundreds of millions, not millions.

4.2 The Received Architecture

Labor law, contract law, banking, and commercial-dispute adjudication each presume a recognized legal subject attached to a jurisdiction that enforces their rights. The relevant populations occupy the negative space of that presumption.

- The employment relation is typically unavailable to them formally: permission-to-work requirements gate access to the employee category, and undocumented work, where it occurs, occurs outside legal recognition.
- The independent-contractor relation is unstable: cannot register a business in many jurisdictions of residence, cannot enforce contracts in local courts that do not recognize the party's standing, cannot reliably serve process across jurisdictional boundaries.
- Banking access is restricted or absent: opening an account typically requires documents of residency and identity that the population does not have or does not have in the form the host jurisdiction recognizes.

- Adjudication has no stable forum: commercial disputes have no local court willing to hear them from a party whose standing is contested.

The humanitarian-economics tradition has documented the cost of this gap. Betts and Collier [2017] argue that displaced populations are systematically prevented from contributing to the economies that host them by an institutional architecture designed for citizens, with the result that the host economy captures little of the productive capacity the displaced bring, and the displaced themselves remain dependent on humanitarian transfers when productive participation would be both possible and preferred. Pritchett [2006] makes the broader argument about labor-mobility friction and the welfare cost of the institutional gap between labor markets in origin and destination economies.

4.3 What the Primitive Does and Does Not Do

The primitive does not resolve statelessness. It does not confer citizenship, restore legal recognition, or create enforcement against a state that denies the person’s standing. It is not a substitute for the political work that addresses those gaps. What it does is structurally narrower:

- (a) It provides a commerce architecture whose precondition is a cryptographic key rather than a recognized civil-legal subjecthood. A wallet does not require a passport to bond; a bonded commitment does not require a host-state court to enforce; a settled exchange does not require a domestic bank to clear.
- (b) It produces a verifiable performance record that travels with the wallet across jurisdictional boundaries. A contributor whose civil-legal subjecthood is contested in jurisdiction *A* accumulates the same on-chain attestation history as one whose status is settled in jurisdiction *B*. The record is portable in a way that conventional employment or contracting records are not.
- (c) It opens a region of productive activity — bilateral, bonded, settled directly between parties — to participants for whom the legal scaffold that conventionally underwrites such activity is unavailable. The set of activities thus opened is bounded by what can be coordinated bilaterally and settled atomically; the bound is non-trivial but not all of productive activity.

These are descriptive properties of the architecture. They do not depend on any humanitarian program or any particular legal reform; they follow from the construction of the primitive.

4.4 Arendt and the Right to Have Rights

Arendt [1951] observed that the fundamental human right is the right to have rights — the right to belong to a political community within which one’s other rights have standing. Her analysis of statelessness named the condition as a civilizational failure rather than an individual misfortune. The person without a polity is not merely deprived of specific rights; they are deprived of the institutional setting in which having-rights is even a coherent claim. The remedy is political, not technological.

The bonded primitive does not grant the right to have rights; it grants, more narrowly, a capacity to have commerce. The two should not be confused. The right to have rights is political and is not within the gift of any technology. The capacity to have commerce is structural and becomes operational once the cryptographic infrastructure is in reach. For populations for whom the first right is denied, the second capacity is non-trivial; for populations for whom the first right is granted, the second capacity is largely subsumed by existing institutions and the primitive is one infrastructure among several. The distinction matters for the audience of this paper

because the primitive’s leverage is asymmetric: where the conventional infrastructure works, it adds modestly; where the conventional infrastructure fails, it adds the architectural condition for participation at all.

5 Operational Constraints

The present paper is positional rather than programmatic. The primitive moves the commerce-access gap from architectural impossibility to operational engineering question; it does not solve the engineering question. We name four constraints that remain.

Key custody. Populations with unreliable access to power, devices, and connectivity face usability barriers at the cryptographic layer that the primitive does not solve. Loss-of-key risk is high in unstable environments, and key-recovery infrastructure for displaced populations is not built. The constraint is real and evolves with the broader infrastructure around the protocol; solutions exist (social recovery, custodial wallets with revocable trust) but each carries trade-offs the population is poorly positioned to evaluate.

Counterparty willingness. A well-resourced buyer in a recognized jurisdiction may decline to bond with a counterparty whose civil-legal standing is contested. The decline is not because of the primitive; it is because of downstream legal exposure in the buyer’s home jurisdiction (sanctions screening, anti-money-laundering obligations, forum-of-enforcement concerns). The primitive does not dissolve this friction. Its effect, where it has one, is to shift the friction from architectural — “commerce with the displaced is impossible” — to discretionary — “some counterparties will, others will not.”

Fiat on-ramp. Converting wallet-denominated value to local currency still runs through existing rails, which remain gated for the population in question. The primitive operates within the crypto-native settlement layer; the bridge between that layer and local purchasing power remains a separate problem with its own architecture and its own gatekeepers. Stablecoin acceptance by local merchants, where it occurs, partially closes this gap; full closure depends on infrastructure outside the primitive.

Collateral access. Bonding requires collateral. A counterparty without starting capital cannot bond. This is a hard constraint on the primitive’s applicability to populations whose displacement is correlated with asset loss. Possible mitigations exist at the assembly tier — sponsor pools, cooperative bonding, assembly-aggregated bonds, microfinance-style group guarantees — but each of these is a construction on top of the primitive, not a property of the primitive itself, and none are yet built at scale.

The cumulative effect of these constraints is that the primitive is not a solved-problem deliverable for displaced populations today. Its present-tense contribution is architectural: commerce-without-recognition is now possible to build, where it was not before, and the constraints that remain are engineering problems with engineering solutions rather than structural features of the legal infrastructure.

6 Conclusion

The bonded settlement primitive does not solve statelessness, and this paper has not argued that it does. It does change the architecture of the commerce-access gap that statelessness creates, by replacing the precondition of recognized civil-legal subjecthood with the precondition of a cryptographic key. The change is not in the political condition of the populations under

discussion; it is in the institutional architecture they face when they attempt to participate in productive activity across jurisdictional boundaries.

The Arendtian distinction is the one to hold: the right to have rights is political, denied by politics, and not within the gift of any technology to restore. The capacity to have commerce is structural, denied by the legal architecture of commerce, and restorable by a settlement primitive whose precondition is cryptographic rather than civil-legal. For populations for whom the first denial is constitutive, the second restoration is non-trivial.

The paper is positional. It identifies a gap, names the architectural change, and acknowledges the operational constraints that remain. The programmatic work — building key-custody infrastructure for refugee populations, designing sponsor pools that lend collateral against verified attestation history, partnering with humanitarian organizations to evaluate where bonded commerce is structurally feasible and where it is not — is for the practitioners who do that work. The contribution of this paper, and its companion at the labor-law side (the labor-law paper), is to make the architectural change legible to scholarship that has not yet recognized it as available.

Acknowledgments

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